

SLOTFORMER: UNSUPERVISED VISUAL DYNAMICS SIMULATION WITH OBJECT-CENTRIC MODELS

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ABSTRACT

Understanding dynamics from visual observations is a challenging problem that requires disentangling individual objects from the scene and learning their interactions. While recent object-centric models can successfully decompose a scene into objects, modeling their dynamics effectively still remains a challenge. We address this problem by introducing SlotFormer – a Transformer-based autoregressive model operating on learned object-centric representations. Given a video clip, our approach reasons over object features to model spatio-temporal relationships and predicts accurate future object states. In this paper, we successfully apply SlotFormer to perform video prediction on datasets with complex object interactions. Moreover, the unsupervised SlotFormer’s dynamics model can be used to improve the performance on supervised downstream tasks, such as Visual Question Answering (VQA), and goal-conditioned planning. Compared to past works on dynamics modeling, our method achieves significantly better long-term synthesis of object dynamics, while retaining high quality visual generation. Besides, SlotFormer enables VQA models to reason about the future without object-level labels, even outperforming counterparts that use ground-truth annotations. Finally, we show its ability to serve as a world model for model-based planning, which is competitive with methods designed specifically for such tasks. Additional results and details are available at our [Website](#).

1 INTRODUCTION

The ability to understand complex systems and interactions between its elements is a key component of intelligent systems. Learning the dynamics of a multi-object systems from visual observations entails capturing *object* instances, their appearance, position and motion, and simulating their spatio-temporal interactions. Both in robotics (Finn et al., 2016; Lee et al., 2018) and computer vision (Shi et al., 2015; Wang et al., 2017), unsupervised learning of dynamics has been a central problem due to its important practical implications. Obtaining a faithful dynamics model of the environment enables future prediction, planning and, crucially, allows to transfer the dynamics knowledge to improve downstream supervised tasks, such as visual reasoning (Chen et al., 2020b; Ding et al., 2021b), planning (Sun et al., 2022) and model-based control (Micheli et al., 2022). Yet, an effective domain-independent approach for unsupervised visual dynamics learning from video remains elusive.

One approach to visual dynamics modeling is to frame it as a prediction problem directly in the pixel space (Shi et al., 2015; Wang et al., 2017; Denton & Fergus, 2018). This paradigm builds on global frame-level representations, and uses dense feature maps of past frames to predict future features. By design, such models are object-agnostic, treating background and foreground modeling as equal. This frequently results in poorly learned object dynamics, producing unrealistic future predictions over longer horizons (Oprea et al., 2020). Another perspective to dynamics learning is through object-centric dynamics models (Kosiorek et al., 2018; van Steenkiste et al., 2018; Kossen et al., 2019). This class of methods first represents a scene as a set of object-centric features (a.k.a. slots), and then learns the interactions among the slots to model scene dynamics. It allows for more natural dynamics modeling and leads to more faithful simulation (Veerapaneni et al., 2020; Zoran et al., 2021). To achieve this goal, earlier object-centric models bake in strong scene (Jiang et al., 2019) or object (Lin et al., 2020) priors in their frameworks, while more recent methods (Kipf et al., 2019; Zoran et al., 2021) learn object interactions purely from data, with the aid of Graph Neural Networks (GNNs) (Battaglia et al., 2018) or Transformers (Vaswani et al., 2017). Yet, these approaches

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SLOTFORMER: 基于对象中心模型的无监督视觉动力学仿真

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摘要

通过视觉观测理解物体动态是一个极具挑战性的任务, 需要从场景中分离出各个独立物体并学习它们之间的交互关系。尽管近期以物体为中心的模型已成功将场景分解为多个物体, 但有效建模其动态行为仍是难题。我们通过提出SlotFormer——一种基于Transformer架构的自回归模型, 该模型基于学习到的物体中心化表征进行运算——来解决这一问题。给定一段视频片段, 我们的方法通过对物体特征进行推理来建模时空关系, 并准确预测未来物体状态。本文成功应用SlotFormer对包含复杂物体交互的数据集进行了视频预测。此外, 无监督训练的SlotFormer动态模型还能提升有监督下游任务 (如视觉问答 VQA 和目标条件规划) 的性能。与以往动态建模研究相比, 我们的方法在保持高质量视觉生成的同时, 实现了显著更优的物体动态长期合成效果。此外, SlotFormer使 VQA 模型能够在无需对象级标签的情况下进行未来预测, 其性能甚至优于使用真实标注数据的同类模型。

最后, 我们证明了该模型能够作为基于模型的规划方法的世界模型, 其性能与专为此类任务设计的方法相当。更多结果与详细信息可访问我们的网站获取。

1 引言

理解复杂系统及其各要素间相互作用的能力是智能系统的关键组成部分。通过视觉观察学习多对象系统的动态特性, 需要捕获对象实例、其外观、位置与运动, 并模拟它们的时空交互关系。无论是在机器人学 (Finn 等人, 2016; Lee 等人, 2018) 还是计算机视觉领域 (Shi等人, 2015; Wang 等人, 2017), 动态无监督学习因其重要的实际应用价值始终是核心研究课题。构建环境的精确动态模型能够实现未来的预测与规划, 并关键地将动态知识迁移至改进下游监督任务, 例如视觉推理 (Chen 等人, 2020b; Ding 等人, 2021b)、路径规划 (Sun 等人, 2022) 以及基于模型的控制 (Micheli 等人, 2022)。然而, 一种有效的、不依赖于结构域的、用于从视频中无监督学习视觉动态的方法至今仍未被实现。

视觉动态建模的一种方法是将其直接视为像素空间中的预测问题 (Shi 等人, 2015; Wang 等人, 2017; Denton & Fergus, 2018)。该范式基于全局帧级表征, 利用历史帧的密集特征图来预测未来特征。从设计上讲, 此类模型具有对象无关性, 将背景与前景建模视为同等重要。这常常导致物体动力学学习效果不佳, 在较长时间跨度内产生不真实的未来预测 (Oprea 等人, 2020)。另一种动态学习视角是通过以物体为中心的动力学模型 (Kosiorrek等人, 2018; van Steenkiste等人, 2018; Kossen等人, 2019)。这类方法首先将场景表示为一组以物体为中心的特征 (即槽位), 然后通过学习槽位间的交互关系来建模场景动态。它能够实现更自然的动力学建模, 并产生更逼真的模拟结果 (Veerapaneni 等人, 2020; Zoran 等人, 2021)。为实现这一目标, 早期以对象为中心的模型在其框架中内置了强大的场景 (Jiang 等人, 2019) 或对象 (Lin 等人, 2020) 先验知识; 而较新的方法 (Kipf 等人, 2019; Zoran 等人, 2021) 则纯粹通过数据学习对象交互, 借助图神经网络 (GNNs) (Battaglia等人, 2018) 或Transformer (Vaswani 等人, 2017)。然而, 这些方法

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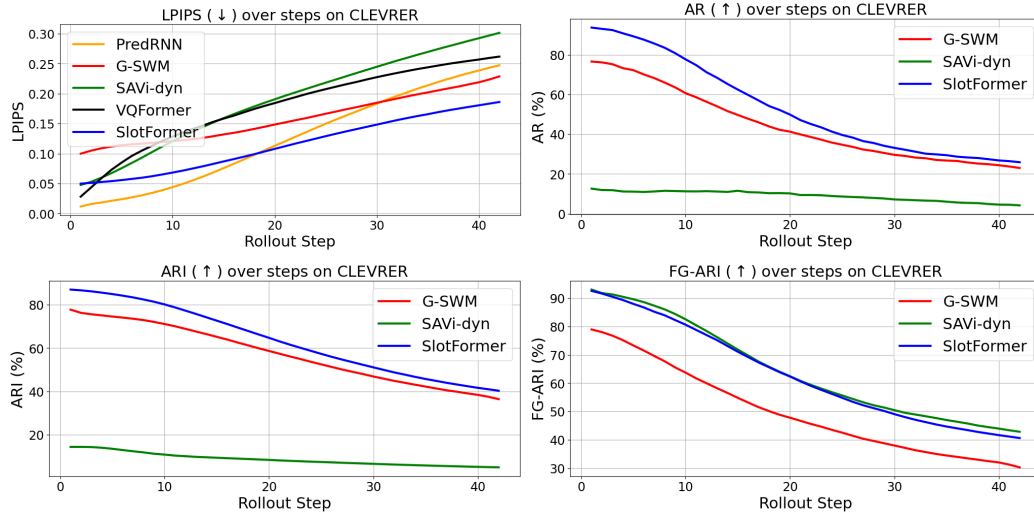


Figure 9: Comparison of the object dynamics of the generated videos at each rollout step on CLEVRER. We report FG-API (left) and FG-mIoU (right) of the segmentation masks.

Method	Predictive	
	per opt. (%)	per ques. (%)
DCL	90.5	82.0
VRDP	91.7	83.8
VRDP [†]	94.5	89.2
Aloe	93.5	87.5
Aloe*	93.1	87.3
Aloe* + Ours	96.5	93.3

Table 9: Predictive VQA on CLEVRER, reporting per-option (per opt.) and per-question (per ques.) accuracy. DCL and VRDP[†] both utilize pre-trained object detectors; * indicates our re-implementation.

Method	Obs. (%)	Dyn. (%)	↑ (%)
Human	74.7	-	-
RPIN	54.3	55.7	+ 1.4
RPIN*	62.8	63.8	+ 1.0
pDEIT-lstm	59.9	60.5	+ 0.6
pDEIT-lstm*	59.2	60.0	+ 0.8
Ours	65.2	67.1	+ 1.9

Table 10: VQA accuracy on Physion. We report the readout accuracy on observation (OBS.) and observation plus rollout (Dyn.) frames. ↑ denotes the improvement brought by the learned dynamics. Methods marked with * are our reproduced results.

PSNR and SSIM are still close to PredRNN and SAVi-dyn, which blurs the moving objects into the background in later frames. This again proves that LPIPS are superior metrics for measuring the generated videos. Besides, G-SWM can also preserve the object identity because it leverages complex priors such as depth to model occlusions. Nevertheless, its simulated dynamics are still worse than our Transformer model. Finally, VQFormer is able to generate sharp images without blurry objects because of its strong VQ-VAE decoder. However, the object properties such as colors are not consistent, and the simulated dynamics are erroneous.

We present a visual result on CLEVRER in Figure 7 (bottom). The objects are smaller in size and have longer term dynamics, making it much harder than OBJ3D. PredRNN and SAVi-dyn still generate blurry objects at later steps. G-SWM sometimes cannot detect objects newly entering the scene because of the limited capacity of its discovery module. In contrast, SlotFormer builds Transformer on SAVi slots, enabling both accurate object detection and precise dynamics modeling. This is also verified by the object-aware metrics AR and mIoU we show in the figures.

See Figure 13 and Figure 14 for more qualitative results in the video prediction task.

VQA. Figure 8 shows a qualitative result on Physion dataset, where SlotFormer successfully synthesizes the contact of the red object and the yellow ground. Note the low quality of the predicted frames is due to the STEVE’s Transformer-based decoder, which is not designed for pixel-space reconstruction⁷. Improving the decoder design is beyond the scope of this paper.

⁷In our experiments, even the reconstruction results of STEVE on Physion videos are of low quality.

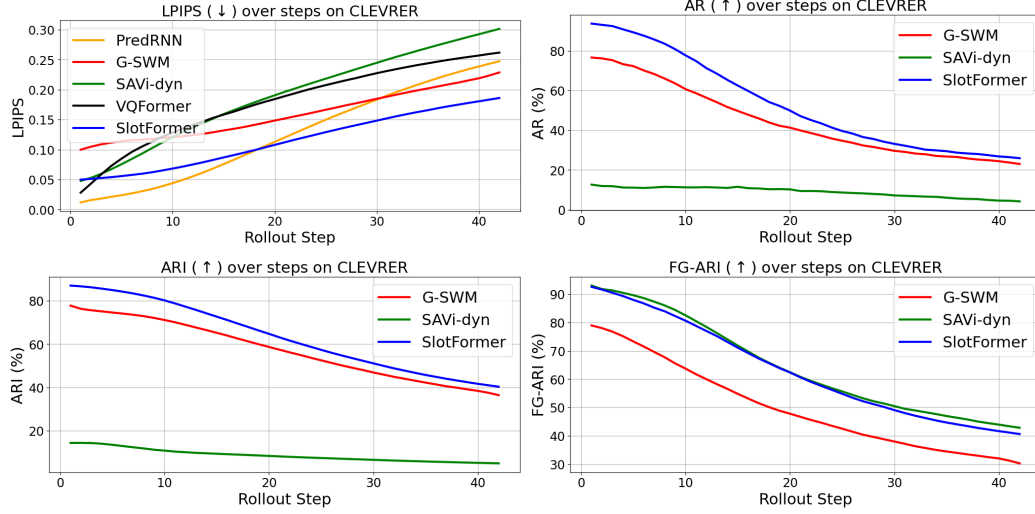


图9：在clever平台各渲染步骤生成视频的对象动态对比。图中展示了分割掩码的FG-API（左）与FG-mIoU（右）指标。

方法	预测性的	
	按选项计算 (%)	按问题 (%)
DCL	90.5	82.0
VRDP	91.7	83.8
VRDP [†]	94.5	89.2
芦荟	93.5	87.5
芦荟*	93.1	87.3
芦荟*+我们的	96.5	93.3

表9：clever模型的预测 VQA，分别报告了每个选项（per opt.）和每个问题（per ques.）的准确率。DCL 和 VRDP[†] 均采用了预训练的目标检测器；*表示我们重新实现的版本。

方法	备注 (%)	动态 (%)	↑ (%)
人	74.7	-	-
RPIN	54.3	55.7	+ 1.4
RPIN*	62.8	63.8	+ 1.0
pDEIT-lstm	59.9	60.5	+ 0.6
pDEIT-lstm*	59.2	60.0	+ 0.8
我们的	65.2	67.1	+ 1.9

表10：Physion数据集上的 VQA 准确率。我们报告了观测帧（OBS.）及观测加动态扩展帧（Dyn.）的读取准确率。↑表示通过学习动态机制带来的改进效果；标有*的方法为我们的重复实验结果。

PSNR 和 SSIM 的性能仍接近PredRNN和SAVi-dyn，这两种方法会将运动物体模糊融入后续帧的背景中。这再次证明 LPIPS 是评估生成视频质量的更优指标。此外，G-SWM 还能保留物体身份特征，因为它利用深度等复杂先验信息来建模遮挡效果。不过，其模拟的动力学表现仍逊于我们的Transformer模型。最后，VQFormer凭借强大的 VQ - VAE 解码器能够生成清晰无模糊物体的图像，但物体属性（如颜色）缺乏一致性，且模拟的动力学过程存在偏差。

我们在图7（底部）展示了clever的可视化结果。该模型生成的对象尺寸更小且动态特征更持久，因此其性能远优于OBJ3D。PredRNN和SAVi-dyn在后续步骤中仍会生成模糊的对象；由于其检测模块容量有限，G-SWM 有时无法识别新进入场景的对象。相比之下，SlotFormer基于SAVi槽位构建Transformer模型，既能实现精准的对象检测，又能进行精确的动力学建模。这一优势也通过图中展示的对象感知指标AR和mIoU得到了验证。

视频预测任务的更多定性结果详见图13和图14。

VQA.图8展示了在Physion数据集上的定性结果：SlotFormer成功合成了红色物体与黄色背景之间的接触细节。需注意，预测帧质量较低是由于采用了基于Transformer的解码器——该解码器并非专为像素空间重建而设计⁷。改进解码器设计不在本文研究范围内。

⁷在我们的实验中，即使使用Physion视频对steve进行重建所得的结果，其质量也较低。

Method	Descriptive	Explanatory		Predictive		Counterfactual		Average
		per opt.	per ques.	per opt.	per ques.	per opt.	per ques.	
Aloc* + Ours	95.17	98.04	94.79	96.50	93.29	90.63	73.78	89.26

Table 11: Accuracy on different questions and average results on CLEVRER. Numbers are in %.

Method	Collide	Contain	Dominoes	Drape	Drop	Link	Roll	Support	Average
SlotFormer	69.3	63.3	55.6	66.7	62.7	69.3	70.7	77.3	67.1

Table 12: Accuracy breakdown for all eight scenarios on Physion. Numbers are in %.

Method	Fold ID										Average
	0	1	2	3	4	5	6	7	8	9	
SlotFormer	83.1	83.2	81.0	81.2	81.2	83.0	82.6	80.0	83.0	81.8	82.0 $\pm$ 1.1

Table 13: AUCESS for all 10 folds on PHYRE.

Method	OBJ3D			CLEVRER			Method	AR $\uparrow$	ARI $\uparrow$	FG-ARI $\uparrow$	FG-mIoU $\uparrow$
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$					
OCVT	31.08	0.88	0.13	27.96	0.87	0.18	OCVT	36.19	51.23	40.87	20.57
Slot-LSTM	32.15	0.90	0.09	29.79	0.88	0.13	Slot-LSTM	48.52	59.58	58.42	27.84
Ours	32.40	0.91	0.08	30.21	0.89	0.11	Ours	53.14	63.45	63.00	29.81

Table 14: Evaluation of visual quality on both datasets.**Table 15:** Evaluation of object dynamics on CLEVRER. All the numbers are in %.

E.2 QUANTITATIVE RESULTS

Video prediction. We show the per-step FG-ARI and FG-mIoU results in Figure 9. The sophisticated priors in G-SWM prevents it from scaling to scenes with multiple objects and complex dynamics. Since SAVi-dyn generates blurry objects, it produces many false positives in the segmentation masks. Instead, SlotFormer preserves the object identity and achieves high scores in both metrics over long rollout steps.

VQA. Table 9 and Table 10 present the complete results of the original and our reproduced performance of baselines, as well as ours on both VQA datasets. We report the performance of our Aloe with SlotFormer model on all four question types of CLEVRER in Table 11. We report the per-scenario accuracy of SlotFormer on Physion rollout setting in Table 12.

Planning. Table 13 shows the AUCESS of SlotFormer for all 10 folds on PHYRE.

E.3 COMPARISON WITH ADDITIONAL BASELINES

In this section, we compare SlotFormer with two additional baselines, namely *OCVT* (Wu et al., 2021) and *Slot-LSTM*, in the video prediction task on OBJ3D and CLEVRER datasets.

OCVT builds Transformers over SPACE (Lin et al., 2019) which applies heavy priors in their framework. In addition, it requires a Hungarian alignment step of slots for loss computation. Therefore, OCVT underperforms G-SWM (Lin et al., 2020) in long-term generation. Since its code is not released, we reproduce it based on SPACE⁸, and adopt its setting in the video prediction task.

As shown in Table 14 and Table 15, OCVT underperforms SlotFormer in both visual quality of videos and accuracy of object dynamics. This is because object slots from SAVi is more powerful than SPACE, and SlotFormer naturally enjoys the temporal alignment of slots.

Slot-LSTM trains a Transformer-LSTM dynamics module from PARTS (Zoran et al., 2021) over the same pre-trained object slots as SlotFormer. We adopt the same Transformer module as SlotFormer, but only feed in slots at a single timestep, thus only modeling the spatial interaction of objects. The Transformer is followed by a per-slot LSTM to learn the temporal dynamics.

As shown in Table 14 and Table 15, SlotFormer outperforms Slot-LSTM in all the metrics, especially on the more challenging CLEVRER dataset. This indicates the importance of joint spatial-temporal reasoning over a larger context window. Despite having the same Transformer module, Slot-LSTM

⁸<https://github.com/zhixuan-lin/SPACE>

方法	描述性的	解释性的		预测性的		反事实的		平均值
		根据选项选择：根据问题确定。		根据选项选择：根据问题确定。		根据选项选择：根据问题确定。		
芦荟*+我们的	95.17	98.04	94.79	96.50	93.29	90.63	73.78	89.26

表11：不同问题的准确率及Clever工具的平均结果。数值以百分比表示。

方法	碰撞	包含：属于	多米诺骨牌	将悬挂	下棋：删除	链式滚筒	支持	平均值	
插槽成型机	69.3	63.3	55.6	66.7	62.7	69.3	70.7	77.3	67.1

表12：Physion平台上所有八种场景的准确率细分数据。数值以百分比（%）表示。

方法	0	1	2	3	4	5	6	7	8	9	平均值
插槽成型机	83.1	83.2	81.0	81.2	81.2	83.0	82.6	80.0	83.0	81.8	82.0 $\pm$ 1.1

表13：在 PHYRE 上所有10个折叠均取得成功。

方法	OBJ3D			CLEVRER			方法AR $\uparrow$ ARI $\uparrow$ FG-ARI $\uparrow$ FG-mIoU $\uparrow$				
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$					
OCVT	31.08	0.88	0.13	27.96	0.87	0.18	36.19	51.23	40.87	20.57	
Slot-LSTM	32.15	0.90	0.09	29.79	0.88	0.13	48.52	59.58	58.42	27.84	
我们的	32.40	0.91	0.08	30.21	0.89	0.11	53.14	63.45	63.00	29.81	

表14：两个数据集的视觉质量评估。

表15：对clever中对象动态特性的评估。所有数值均以百分比表示。

E.2 定量结果

视频预测。图9展示了每一步的FG-ARI和FG-mIoU结果。G- SWM 中复杂的先验知识使其难以适用于包含多个物体和复杂动态的场景。由于SAVi-dyn会生成模糊物体，因此会在分割掩码中产生大量误报；相比之下，SlotFormer能保留物体的身份信息，并在长时间的预测过程中两项指标均取得高分。

VQA.表9和表10展示了原始基线模型及我们重新实现的基线模型在两个 VQA 数据集上的完整性能结果。我们在表11中报告了Aloe与SlotFormer模型在Clever数据集所有四种问题类型上的表现；表12则展示了SlotFormer在Physion部署场景下的各场景准确率。

规划。表13展示了SlotFormer在 PHYRE 数据集上所有10个交叉验证折上的表现。

E.3 与其他基线数据的比较

在本节中，我们将在OBJ3D和CLEVERER数据集上的视频预测任务中，将SlotFormer与另外两种基线方法——OCVT（Wu 等人，2021）和Slot-LSTM——进行比较。

OCVT在空间维度上构建Transformer（Lin 等人，2019），其框架中采用了强烈的先验假设。此外，该方法需要进行匈牙利语槽位对齐步骤以计算损失函数。因此，在长期生成任务中，OCVT的表现逊于G- SWM（Lin 等人，2020）。由于其代码未公开，我们基于空间维度⁸对其进行了复现，并将其设置应用于视频预测任务。如表14和表15所示，OCVT在视频视觉质量与物体动态准确性两方面均不及SlotFormer。这是因为SAVi提供的物体槽位比空间维度更强大，而SlotFormer天然具备槽位的时间对齐优势。

Slot-LSTM基于与SlotFormer相同的预训练对象槽位数据（Zoran 等人，2021），训练一个Transformer-LSTM 动态模块。我们采用与SlotFormer相同的Transformer模块，但仅输入单一时步长的对象槽位信息，从而仅建模对象间的空间交互关系。该Transformer模块后接一个针对每个槽位的 LSTM 网络，用于学习时间动态特征。

如表14和表15所示，SlotFormer在所有评估指标上均优于Slot-LSTM，尤其在更具挑战性的clever数据集上表现更优。这表明在更大的上下文窗口中进行联合时空推理的重要性。尽管两者采用相同的Transformer模块，Slot-LSTM

⁸ <https://github.com/zhixuan-lin/space>

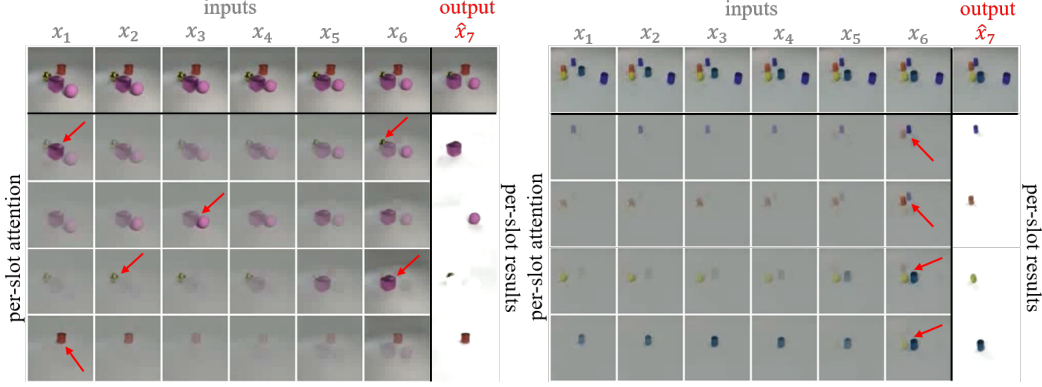


Figure 10: Example attention map visualization on OBJ3D (left) and CLEVRER (right). Our model takes in slots from $\{x_i\}_{i=1}^6$ (column 1-6) to predict slots of $\hat{x}_7$ (column 7). We show images at the first row and the per-slot future reconstructions at the rightmost column. The body of the table shows the per-slot attention of SlotFormer when predicting $\hat{\mathcal{S}}_7$, with the arrows pointing at the regions of high importance for predicting the future slot in the same row. Zoom in for better viewing.

limits the context window of its recurrent module to only a single timestep. Therefore, it still generates videos with blurry objects and inconsistent dynamics over the long horizon.

E.4 ATTENTION ANALYSIS

In this section, we analyze the visual cues in the input frames that SlotFormer utilizes to make future predictions. We do so by visualizing the attention map from the last self-attention layer in the transformer $\mathcal{T}$. More precisely, given the last T encoded frames $\{\mathcal{S}_t\}_{t=1}^T$, we are predicting the future slots $\hat{\mathcal{S}}_{T+1}$. Denote the attention scores from $\hat{\mathcal{S}}_{T+1}^i$ to $\mathcal{S}_t^j$ as $a_{t,j}^i$, where $i, j \in [1, N]$ and N is the number of slots. At each timestep t and for each future slot i , we obtain spatial attention maps o_t^i over input frames x_i as a weighted combination of the slot reconstructions as follows:

$$o_t^i = \sum_{j=1}^N a_{t,j}^i \cdot (m_t^j \odot y_t^j), \quad (8)$$

which indicates the regions of x_t SlotFormer attends upon when predicting $\hat{\mathcal{S}}_{T+1}^i$.

Figure 10 (left) presents one example from OBJ3D, where the purple cube just collided with the purple sphere, and is about to hit the yellow sphere. When predicting the purple cube, the model focuses on the past collision event in $\{x_i\}_{i=1}^4$, and highlights the yellow sphere in x_6 . For the purple sphere, the Transformer only looks at the purple cube because it will not hit the yellow sphere. Since the yellow sphere becomes heavily occluded in x_6 , SlotFormer attends to earlier frames, while predicting its future motion based on the purple cube. This indicates that SlotFormer can handle occlusions or disappearing of objects during burn-in frames by attending to other timesteps where the objects are visible, and using that information to infer the properties and motion of objects. Finally, the red cylinder merely looks at itself because it is not involved in the collisions.

Figure 10 (right) illustrates one example from CLEVRER. We only analyze the left side of the images since there is no object interaction in the right part. There are two collision events (the purple cylinder hitting the orange cylinder, and the yellow sphere hitting the blue cube) happening in x_7 , and SlotFormer successfully captures their interactions in the attention maps. In general, we found the attention maps in CLEVRER less clear than those in OBJ3D, due to the smaller object size. Nevertheless, the Transformer can still detect correct cues to reason their future motion.

E.5 ABLATION STUDY

Figure 11 shows the effect of burn-in and rollout length on SlotFormer’s performance as line plot for better clarity. On OBJ3D, the LPIPS first improves as we use more burn-in frames, and then degrades after reaching a peak at $T = 6$. We do not ablate the rollout length as it is fixed according to the evaluation setting. On Physion, the accuracy gain increases consistently with more burn-in frames as it provides more context information. Therefore, we choose the maximum length $T = 15$ according to the number of observed frames available at test time. Finally, the accuracy grows as we use more rollout frames during training, and plateaus after $K = 10$.

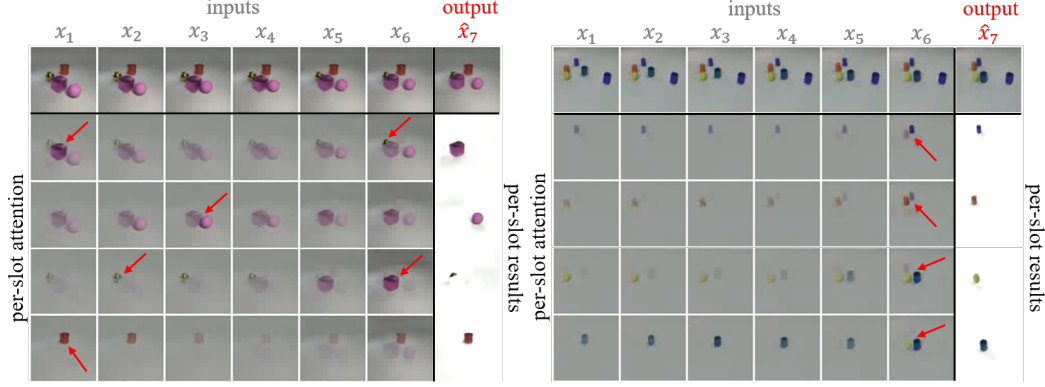


图 10: 在 OBJ3D (左) 和 clevrer (右) 上注意力图可视化的示例。我们的模型接收来自 $\{x_i\}_{i=1}^6$ (第 1-6 列) 的槽位数据, 以预测 $\hat{x}_7$ (第 7 列) 的槽位。我们在第一行展示图像, 并在最右侧列显示每个槽位的未来重构结果。表格主体展示了 SlotFormer 在预测 $\hat{x}_7$ 时的逐槽位注意力分布, 箭头指向同一行中对预测未来槽位至关重要的区域。可放大查看以获得更清晰效果。

该方法将其循环模块的上下文窗口限制为仅一个时间步长。因此, 生成的视频中物体仍会出现模糊现象, 且在长时间跨度内动态表现不一致。

E.4 注意力分析

在本节中, 我们分析了 SlotFormer 用于进行未来预测的输入帧中的视觉线索。我们通过可视化 Transformer T 中最后一个自注意力层生成的注意力图来实现这一分析。更具体地说, 给定最后的 T 编码帧 $\{S_t\}_{t=1}^T$, we are predicting the 未来时隙 $\hat{S}_{T+1}$ 。将从 $\hat{S}_{T+1}$ 以及 s_{jt} 的注意力分数记为 $a_{it,j}$, 其中 $i, j \in [1, N]$, 且 N 表示时隙数量。在每个时间步 t 及每个未来时隙 i 上, 我们通过以下方式对输入帧 x_i 计算空间注意力图 oit , 该图是各时隙重构结果的加权组合:

$$o_t^i = \sum_{j=1}^N a_{t,j}^i \cdot (m_t^j \odot y_t^j), \quad (8)$$

这表明了 x_t SlotFormer 在预测 $\hat{S}_{T+1}$ 时所关注的区域。

图 10 (左) 展示了 OBJ3D 中的一个示例: 紫色立方体刚刚与紫色球体发生碰撞, 即将撞击黄色球体。在预测紫色立方体时, 模型聚焦于历史碰撞事件 $\{x_i\}_{i=4}^6$, 并在 x_6 中标记出黄色球体; 而对于紫色球体, Transformer 仅关注该立方体本身, 因为它不会撞击黄色球体。由于黄色球体在 x_6 中被严重遮挡, SlotFormer 会关注更早的时间帧, 并基于紫色立方体预测其未来运动。这表明 SlotFormer 能够通过关注物体可见的其他时间步, 利用这些信息推断物体的属性和运动, 从而有效处理预热帧中物体的遮挡或消失现象。最后, 红色圆柱体仅关注自身状态, 因为它未参与任何碰撞过程。

图 10 (右) 展示了来自 clevrer 数据集的一个示例。我们仅分析图像的左侧区域, 因为右侧不存在物体交互。在 x_7 场景中发生了两起碰撞事件 (紫色圆柱体撞击橙色圆柱体, 黄色球体撞击蓝色立方体), SlotFormer 成功在注意力图中捕捉到了这些交互。总体而言, 由于 clevrer 数据集中物体尺寸较小, 其注意力图的清晰度低于 OBJ3D 数据集。尽管如此, Transformer 仍能准确识别出用于推断物体未来运动的有效线索。

E.5 消融研究

图 11 以折线图形式清晰展示了预热阶段和展开长度对 SlotFormer 性能的影响。在 Obj3D 数据集上, LPIPS 值首先随着预热帧数量的增加而提升, 随后在 $T=6$ 时达到峰值后开始下降; 由于展开长度根据评估设置固定不变, 本研究未对其进行消融实验。在 Physion 数据集上, 精度增益随预热帧数量的增加持续上升, 因为预热帧能提供更丰富的上下文信息。因此, 我们根据测试时可用帧数选择了最大展开长度 $T=15$ 。最终, 随着训练过程中展开帧数量的增加, 精度持续提升, 并在 $K=10$ 时趋于平稳。

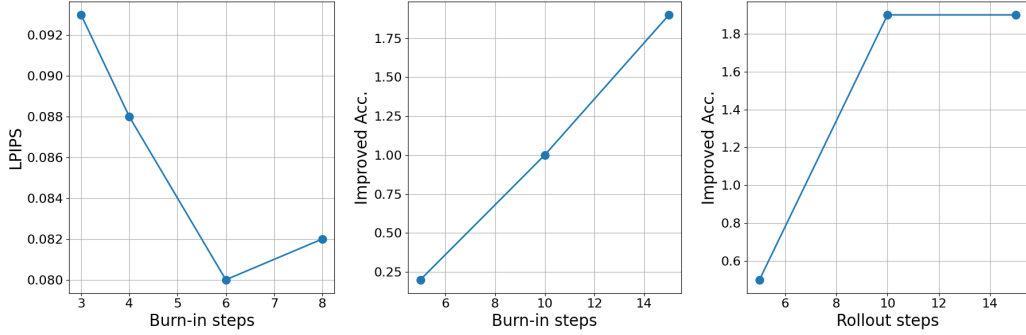


Figure 11: Ablation study on burn-in and rollout length of SlotFormer. We show the LPIPS of generated videos on OBJ3D (left), and the improved VQA accuracy by rollout on Physion (middle, right).

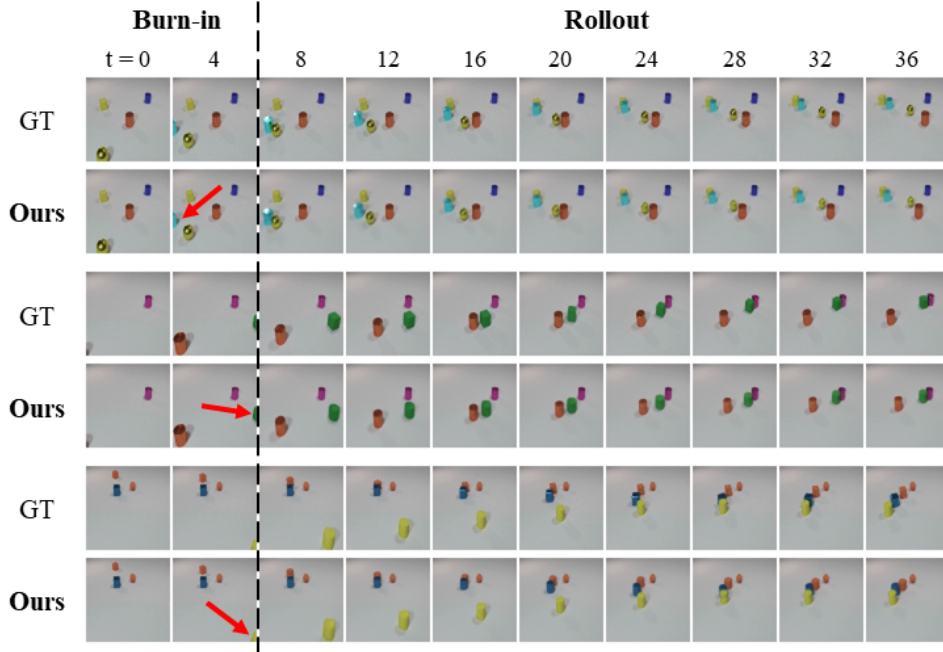


Figure 12: Videos generated by SlotFormer on CLEVRER where some objects are not visible at the initial frame, but enters the scene during burn-in frames (marked by red arrows). SAVi is able to detect these new objects, and SlotFormer can still simulate accurate future dynamics for these objects.

E.6 NEW OBJECTS DURING BURN-IN FRAMES

One concern regarding SlotFormer is that, if some objects are not visible at the first timestep, but enters the scene during burn-in frames, can our model still be able to simulate their dynamics? Figure 12 shows a few examples with new objects appearing during burn-in steps on the CLEVRER dataset. Since the object-centric model SAVi is able to detect these new objects, SlotFormer can still reason their dynamics based on frames where they are visible, and simulate accurate future states. This is an important property of SlotFormer as disappearing and (re-)appearing of objects are common in real world videos.

F LIMITATIONS AND FUTURE WORKS

Limitations. SlotFormer currently builds upon pretrained object-centric models. This family of methods still fail to scale up to real world data⁹, preventing our application to real world videos as well. Besides, the two-stage training strategy harms the model performance at the early rollout

⁹STEVE (Singh et al., 2022) can work on real world videos such as traffics, where ground usually shares the same color, and looks distinct from the vehicles. Also, STEVE’s Transformer-based slot decoder cannot generate images of high visual quality on complex datasets.

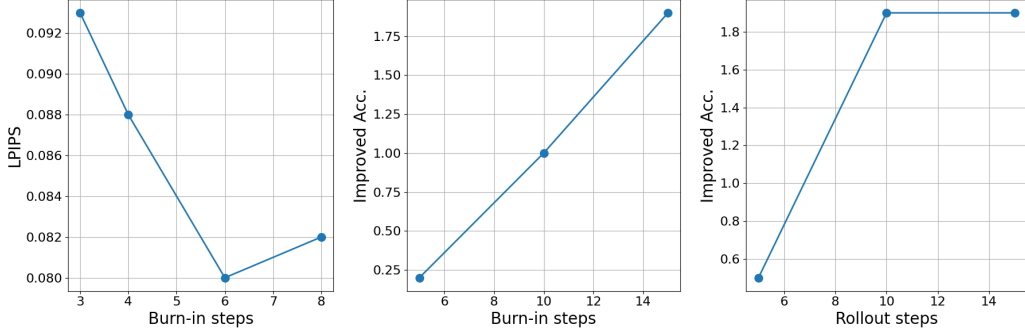


图11: SlotFormer的烧录时间与渲染长度消融实验。图中展示了在OBJ3D上生成视频的 LPIPS（左图），以及通过Phys-ion渲染技术提升的 VQA 精度（中图、右图）。

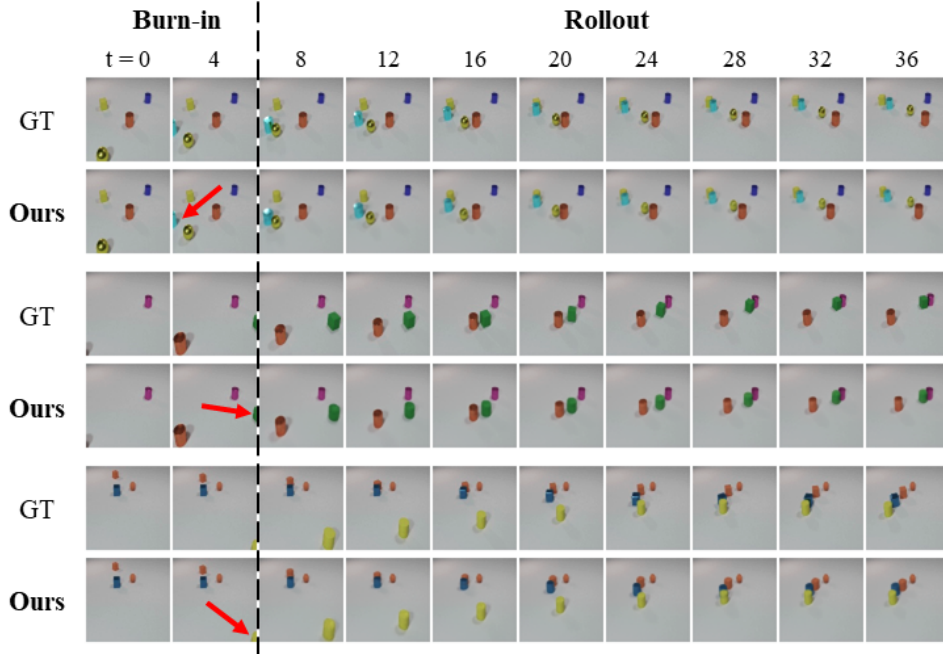


图12: SlotFormer在clever数据集上生成的视频显示：部分物体在初始帧中不可见，但在烧录帧期间进入场景（由红色箭头标示）。SAVi能够检测到这些新物体，而SlotFormer仍能为这些物体模拟出精确的未来动态。

E.6 烧录帧期间的新对象

关于SlotFormer的一个关注点是：如果某些物体在初始时间步未可见，但在烧录帧期间进入场景，我们的模型是否仍能模拟其动态行为？图12展示了Clevrer数据集上烧录阶段出现新物体的若干示例。由于以物体为中心的SAVi模型能够检测到这些新物体，SlotFormer仍能基于其可见帧推断其动态行为，并准确模拟未来状态。这一特性对SlotFormer至关重要，因为在现实世界视频中，物体的消失与（重新）出现现象十分常见。

F 局限性与未来工作

局限性。 SlotFormer目前基于预训练的以对象为中心的模型构建。这类方法仍无法扩展到现实世界数据⁹，因此我们也无法将其应用于现实世界的视频。此外，两阶段训练策略会损害模型在早期部署阶段的性能。

⁹steve (Singh 等人, 2022) 能够处理现实世界的视频（如交通场景），其中地面通常与车辆颜色相同，且与车辆明显区分。此外，steve基于Transformer的时隙解码器在复杂数据集上无法生成高视觉质量的图像。

steps as shown in Figure 2. It is interesting to explore joint training of the base object-centric model and the Transformer dynamics module, which could potentially benefit the performance of both models. Finally, current SlotFormer model works in a deterministic manner, and thus cannot model the uncertainty of future dynamics, which is common in real world videos.

Future Works. We only experiment on unconditional future prediction in this paper. In the future, we plan to extend SlotFormer to conditional generation tasks, such as action-conditioned generation as done in Zoran et al. (2021). Recent works have shown success in this direction by converting conditional inputs to tokens and feeding them to the Transformer (Ren & Wang, 2022; Tevet et al., 2022). Another direction is to simplify the training process by learning scene decomposition and temporal dynamics jointly. This may allow the object-centric model to leverage long-term motion cues for unsupervised object discovery (Yang et al., 2021). Finally, it is important to enable SlotFormer to learn the multi-modality of future dynamics for stochastic video prediction. This is key to modeling real world videos faithfully (Oprea et al., 2020).

	t = 8	12	16	20	24	28	32	36	40	44				
GT												PSNR↑	SSIM↑	LPIPS↓
PredRNN												32.95	0.90	0.12
G-SWM												28.24	0.82	0.19
SAVi-dyn												31.75	0.89	0.13
VQFormer												23.43	0.74	0.28
Ours												31.93	0.91	0.08

	t = 8	12	16	20	24	28	32	36	40	44				
GT												PSNR↑	SSIM↑	LPIPS↓
PredRNN												33.99	0.92	0.08
G-SWM												29.68	0.87	0.09
SAVi-dyn												32.65	0.92	0.10
VQFormer												26.95	0.84	0.13
Ours												31.63	0.92	0.05

Figure 13: More qualitative results on OBJ3D.

具体步骤如图2所示。值得探讨的是对基础对象中心模型与Transformer动态模块进行联合训练，这有望提升两个模型的性能。最后，当前SlotFormer模型采用确定性机制运行，因此无法模拟现实视频中常见的未来动态不确定性。

Future Works. 本文仅针对无条件未来预测进行实验。未来，我们计划将SlotFormer扩展至条件生成任务，例如Zoran 等人（2021）所实现的动作条件生成任务。近期研究已在此方向取得成功，方法是将条件输入转换为令牌并输入Transformer（Ren & Wang, 2022; Tevet 等人, 2022）。另一方向是通过联合学习场景分解与时间动态来简化训练过程，这可能使以对象为中心的模型能够利用长期运动线索进行无监督对象发现（Yang 等人, 2021）。最后，关键是要使SlotFormer能够学习未来动态的多模态特征，以实现随机视频预测——这是准确建模现实世界视频的核心（Oprea 等人, 2020）。

	t = 8	12	16	20	24	28	32	36	40	44				
GT												PSNR↑	SSIM↑	LPIPS↓
PredRNN												32.95	0.90	0.12
G-SWM												28.24	0.82	0.19
SAVi-dyn												31.75	0.89	0.13
VQFormer												23.43	0.74	0.28
Ours												31.93	0.91	0.08

	t = 8	12	16	20	24	28	32	36	40	44				
GT												PSNR↑	SSIM↑	LPIPS↓
PredRNN												33.99	0.92	0.08
G-SWM												29.68	0.87	0.09
SAVi-dyn												32.65	0.92	0.10
VQFormer												26.95	0.84	0.13
Ours												31.63	0.92	0.05

图 13: 关于 OBJ3D 的更多定性结果。

	t = 8	12	16	20	24	28	32	36	40	44			
GT											LPIPS↓	AR↑	mIoU↑
PredRNN											0.11	-	-
G-SWM											0.15	43.85	25.05
SAVi-dyn											0.15	4.76	20.37
Ours											0.04	70.04	36.73
GT											LPIPS↓	AR↑	mIoU↑
PredRNN											0.16	-	-
G-SWM											0.20	29.76	19.81
SAVi-dyn											0.26	11.90	8.08
Ours											0.08	77.98	39.93

Figure 14: More qualitative results on CLEVRER.

	t = 8	12	16	20	24	28	32	36	40	44			
GT											LPIPS↓	AR↑	mIoU↑
PredRNN											0.11	-	-
G-SWM											0.15	43.85	25.05
SAVi-dyn											0.15	4.76	20.37
Ours											0.04	70.04	36.73
GT											LPIPS↓	AR↑	mIoU↑
PredRNN											0.16	-	-
G-SWM											0.20	29.76	19.81
SAVi-dyn											0.26	11.90	8.08
Ours											0.08	77.98	39.93

图 14: 关于 clevrer 的更多定性结果。